Chapter 14: Periodontics & Periodontology

Comprehensive NBDHE Board Exam Review — Questions, Answers & Rationales (30 Questions)

Question 1

What is the name of the anatomical delineation that separates the attached gingival from the alveolar mucosa?

a. Gingival groove
b. Gingiva margin
c. Junctional epithelium
d. Mucogingival junction

Correct Answer: D — Mucogingival junction

Clinical Rationale: Mucogingival junction A. Located 1 to 1.5 mm apical to the gingival margin at the base of the gingival sulcus B. Located at or approximately 0.5 mm coronal to the cemento-enamel junction (CEJ) C. Nonkeratinized epithelium that surrounds and attaches to the tooth on one side and attaches on the other side to the gingival connective tissue D. Mucogingival junction

Question 2

Which region in the mouth is the attached gingiva the narrowest?

a. Mandibular anterior facial
b. Mandibular premolar facial
c. Maxillary anterior lingual
d. Maxillary molar facial

Correct Answer: B — Mandibular premolar facial

Clinical Rationale: Mandibular premolar facial A. Not the narrowest area of attached gingiva B. Mandibular premolar facial C. Is not measured on the palate because it cannot be clinically distinguished from the palatal mucosa D. Typically the widest area of attached gingiva

Question 3

Which part of the periodontium provides a route for fluid and cells can pass from the connective tissue into the sulcus and bacteria and bacteria products to pass from the sulcus into the connective tissue?

a. Gingival sulcus
b. Junctional epithelium
c. Mucogingival junction
d. Periodontal ligament

Correct Answer: B — Junctional epithelium

Clinical Rationale: Junctional epithelium A. Space formed by the tooth and the sulcular epithelium B. Junctional epithelium C. The anatomical delineation that separates the attached gingival from the alveolar mucosa D. Connective tissue fibers that attach the tooth to the alveolar bone

Question 4

What is the length of the junctional epithelium?

a. 0.14 to 0.24 mm
b. 0.25 to 1.35 mm
c. 1.36 to 1.75 mm
d. 1.76 to 2.00 mm

Correct Answer: B — 0.25 to 1.35 mm
Clinical Rationale: 0.24 to 1.35mm A. Too short of length B. 0.25 to 1.35mm C. Too long of length D. Too long of length

Question 5

What is the predominant type of cell of the connective tissue?

a. Fibroblast
b. Macrophage
c. Lymphocyte
d. Mast

Correct Answer: A — Fibroblast
Clinical Rationale: Fibroblast A. Fibroblast B. Immune system cell C. Immune system cell D. Immune system cell

Question 6

What is the term for an isolated area of exposed root on the facial or lingual surface of a tooth which leaves the root directly in contact with gingiva or alveolar mucosa?

a. Bifurcation
b. Dehiscence
c. Exophytic exposure
d. Fenestration

Correct Answer: D — Fenestration
Clinical Rationale: Fenestration A. Description of bone loss between two roots B. Lack of alveolar cortical plate leaving the root exposed C. External growth from the surface of the organ D. Fenestration

Question 7

Which of the following is a calcium-channel blocker that has the potential to cause gingival enlargement?

a. Bisoprolol
b. Cyclosporine
c. Nifedipine
d. Phenytoin

Correct Answer: C — Nifedipine
Clinical Rationale: Nifedipine A. Beta blocker drug B. Immunosuppressive drug C. Nifedipine D. Anti-seizure drug

Question 8

What anatomical structure does the contour of the alveolar bone follow?

a. Cemento-enamel junction
b. Free gingival margin
c. Junctional epithelium
d. Mucogingival junction

Correct Answer: A — Cemento-enamel junction
Clinical Rationale: Cementoenamel junction A. Cementoenamel junction B. Area lo located at or approximately 0.5 mm coronal to the cemento-enamel junction (CEJ) C. Nonkeratinized epithelium that surrounds and attaches to the tooth on one side and attaches on the other side to the gingival connective tissue D. Anatomical delineation that separates the attached gingival from the alveolar mucosa?

Question 9

How far apical is the alveolar crest from the cemento-enamel junction?

a. 0.5 to 1.4 mm
b. 1.5 to 2.0 mm
c. 2.1 to 2.7 mm
d. 2.8 to 3.2 mm

Correct Answer: B — 1.5 to 2.0 mm
Clinical Rationale: 1.5 to 2.0 mm A. Incorrect answer – too short B. 1.5 to 2.0 mm C. Incorrect answer – too long D. Incorrect answer – too long

Question 10

Absence of attachment loss is one feature of biofilm-induced gingivitis. What is the other clinical feature?

a. Bleeding upon probing
b. Bone loss
c. Cervical biofilm
d. Inflammation

Correct Answer: B — Bone loss
Clinical Rationale: Bone loss A. Clinical sign of inflammation B. Bone loss C. Cause of inflammation D. Immune response

Question 11

Which of the following is a major lifestyle risk factor for periodontal disease?

a. Age
b. Large embrasure spaces
c. Orthodontic treatment
d. Smoking

Correct Answer: D — Smoking
Clinical Rationale: Smoking A. Not a lifestyle factor B. Not a lifestyle factor C. Not a lifestyle factor D. Smoking

Question 12

Drug-induced gingival overgrowth is most often observed in which area of the mouth?

a. Anterior
b. Interproximal
c. Posterior
d. All areas

Correct Answer: A — Anterior
Clinical Rationale: Anterior a. Anterior b. Interproximal c. Posterior d. All areas

Question 13

Nonbiofilm-induced gingivitis can be caused by a lack of which vitamin?

a. C
b. D
c. E
d. K

Correct Answer: A — C
Clinical Rationale: Vitamin C (ascorbic acid) deficiency can cause scurvy-related nonbiofilm-induced gingival bleeding and ulceration due to impaired collagen synthesis in capillary walls and gingival connective tissue.

Question 14

What is the most accurate measure of severity of periodontal disease?

a. Amount of bleeding upon probing
b. Amount of subgingival calculus
c. Clinical attachment level
d. Percent of plaque

Correct Answer: C — Clinical attachment level
Clinical Rationale: Clinical attachment level A. widely accepted as an indicator of gingival inflammation and is a poor predictor of attachment loss B. Calculus plays a secondary role in the etiology of periodontal diseases by serving as a plaque-retentive factor C. Clinical attachment level D. Assessment of the patient’s homecare but not indicative of amount of periodontal destruction

Question 15

Which periodontal disease classification is marked by crater-like depressions at the crest of the interdental papilla, interproximal necrosis and severe bleeding?

a. Biofilm-induced gingivitis
b. Drug-induced gingival enlargement
c. Necrotizing periodontal disease
d. Periodontal abscess

Correct Answer: C — Necrotizing periodontal disease

Clinical Rationale: Necrotizing periodontal disease A. Inflammation of the gingiva resulting from biofilm at the gingival margin B. Enlargement of the gingiva C. Necrotizing periodontal disease D. Localized, purulent area of inflammation within periodontal tissue

Question 16

What is the most common region for a pericoronal abscess to occur?

a. Mandibular incisor
b. Mandibular bicuspid
c. Maxillary anterior
d. Third molar

Correct Answer: D — Third molar

Clinical Rationale: Third molar A. Incorrect answer B. Incorrect answer C. Incorrect answer D. Third-molar

Question 17

Which type of cell is the earliest responder in the inflammatory defense during stage I gingivitis?

a. Dendritic cell
b. Leukocyte
c. Lymphocyte
d. Neutrophil

Correct Answer: D — Neutrophil

Clinical Rationale: Neutrophil A. Involved in the adaptive immune response B. White blood cells C. Type of white blood cell D. Neutrophil

Question 18

What is the predominate type of bacteria in apical portions of subgingival-attached plaque?

a. Gram-negative filaments
b. Gram-negative rods
c. Gram-positive cocci
d. Gram-positive rods

Correct Answer: B — Gram-negative rods

Clinical Rationale: Gram negative rods A. Can be present but are not the predominant bacteria B. Gram negative rods C. Usually seen in supragingival plaque D. Usually seen in supragingival plaque

Question 19

What factor does calculus play in the etiology of periodontal diseases?

a. Causative
b. Inflammatory initiating
c. Plaque retentive
d. Nonfactor

Correct Answer: C — Plaque retentive

Clinical Rationale: Plaque-retentive a. Bacterial plaque is the causative factor b. Endotoxins initiate the inflammatory process c. Plaque-retentive d. Calculus is contributing factor in the pathogenesis of periodontal disease

Question 20

Which local factor might contribute to the prognosis of the patient with periodontitis?

a. Malocclusion
b. Presence of systemic disease
c. Patient’s HbA1c level
d. High blood pressure

Correct Answer: A — Malocclusion

Clinical Rationale: Malocculsion A. Malocclusion B. Is not a local factor C. Is not a local factor D. Is not a local factor

Question 21

“To treat and manage established periodontal disease and to create conditions conducive to health” defines which periodontal treatment?

a. Gingival curettage
b. Nonsurgical periodontal therapy
c. Regenerative surgery
d. Resective surgery

Correct Answer: B — Nonsurgical periodontal therapy

Clinical Rationale: Nonsurgical periodontal therapy A. Gingiva a procedure to remove the ulcerated, chronically inflamed tissue lining a periodontal pocket; evidence fails to support the efficacy of this procedure B. Nonsurgical periodontal therapy C. Surgical procedure D. Surgical procedure

Question 22

Success rates in treating early to moderate periodontitis most often occur in which initial pocket depths?

a. 4 to 6 mm
b. 7 to 9 mm
c. 10 to 12 mm
d. Greater than 12 mm

Correct Answer: A — 4 to 6 mm

Clinical Rationale: 4-6 mm A. 4-6 mm B. Less predictable success rate C. Less predictable success rate D. Less predictable success rate

Question 23

What length of time is considered appropriate for resolution of inflammation and periodontal healing after nonsurgical periodontal therapy?

a. 1 to 3 weeks
b. 4 to 6 weeks
c. 7 to 9 weeks
d. 10 to 12 weeks

Correct Answer: B — 4 to 6 weeks

Clinical Rationale: 4-6 weeks A. 4 to 6 weeks is considered appropriate for the resolution of inflammation and periodontal tissue healing B. 4-6 weeks C. 4 to 6 weeks is considered appropriate for the resolution of inflammation and periodontal tissue healing D. 4 to 6 weeks is considered appropriate for the resolution of inflammation and periodontal tissue healing

Question 24

What is the first step when removing nonabsorbable sutures?

a. Insert the scissors under the suture and cut the suture material

b. Cleanse the wound site

c. Grasp the knotted end with cotton pliers and gently pull it away from the tissue

d. Pull the knotted end so that the material is removed from the tissue

Correct Answer: C — Grasp the knotted end with cotton pliers and gently pull it away from the tissue

Clinical Rationale: Grasp the knotted end of the suture with cotton pliers and gently pull it away from the tissue A. The second step of the procedure B. The last step of the procedure C. Grasp the knotted end with cotton pliers and gently pull it away from the tissue D. The third step of the procedure

Question 25

What is the first step when removing periodontal dressing?

a. Gently cleanse the area with warm water or a wet cotton tip

b. Tease the dressing away from the teeth using a curette or cotton pliers

c. Probe the sulcular area

d. Assess the tissue for healing

Correct Answer: B — Tease the dressing away from the teeth using a curette or cotton pliers

Clinical Rationale: Tease the dressing away from the teeth using a curette or cotton pliers A. The second step in removing periodontal dressing B. Tease the dressing away from the teeth using a curette or cotton pliers C. This is a contraindicated procedure during periodontal dressing removal D. The final step in removing periodontal dressing

Question 26

What is the name for the adaptation of keratinized or nonkeratinized epithelium to the abutment cylinder of an implant?

a. Periodontal ligament

b. Osseo-integration

c. Perimucosal seal

d. Perio-implantitis

Correct Answer: C — Perimucosal seal

Clinical Rationale: Perimucosal seal A. Is not part of the implant-tissue interface B. The contact established between normal and remodeled bone and the implant surface without the interposition of connective tissue C. Perimucosal seal D. A periodontitis-like disease process that can affect dental implants

Question 27

What is the minimum success rate of implants at 5 years?

a. 85%
b. 90%
c. 95%
d. 100%

Correct Answer: A — 85%
Clinical Rationale: 85% A. 85% B. Incorrect answer C. Incorrect answer D. Incorrect answer

Question 28

What is the typical amount of alveolar bone loss around implants during the first year after placement?

a. 0 mm
b. 1 to 2 mm
c. 3 to 4 mm
d. 5 to 6 mm

Correct Answer: B — 1 to 2 mm
Clinical Rationale: 1-2 mm a. Initial loss of 1 to 2 mm of alveolar bone during the first year after placement b. 1-2 mm c. Initial loss of 1 to 2 mm of alveolar bone during the first year after placement d. Initial loss of 1 to 2 mm of alveolar bone during the first year after placement

Question 29

What is the name of the procedure which prevents the epithelium from growing downward into the bone-growing surgical site?

a. Autografts
b. Allografts
c. Gingivoplasty
d. Guided tissue regeneration

Correct Answer: D — Guided tissue regeneration
Clinical Rationale: Guided tissue regeneration A. Type of bone graft using bone from the same patient B. Type of bone graft using bone from genetically dissimilar individuals within the same species (e.g., processed human cadaver bone that is freeze-dried or demineralized) C. Reshaping of the gingiva to obtain a physiologic form similar to that characteristic of healthy tissue D. Guided tissue regeneration

Question 30

Which type of fluoride has an antimicrobial mechanism of action?

a. Acidulated phosphate
b. Potassium nitrate
c. Sodium
d. Stannous

Correct Answer: D — Stannous

Clinical Rationale: Stannous A. Type of fluoride but does not have antimicrobial affect B. Not a type of fluoride C. Type of fluoride but does not have antimicrobial affect D. Stannous